Companion
to

ChatGPT Teaches TikZ

Chapter 4: Line Style (B)

Contents

4.1 What This Companion Adds

Chapter 4 separated geometry from appearance. Coordinates and paths determine where a line goes; line-style options determine how it looks. This Companion develops a restrained system for width and pattern.

The examples remain geometrically simple. That makes it possible to see what each option changes and, equally important, what it does not change.

Design

Use differences in line style to express differences in meaning. If every line is emphasized, emphasis disappears.

4.2 A Line-Style Vocabulary

Option	**Purpose**
`thick`	Adds moderate emphasis.
`very thick`	Adds stronger emphasis.
`line width=1.2pt`	Sets a precise width.
`dashed`	Uses a standard dash pattern.
`dotted`	Uses a standard dotted pattern.
`dash pattern=`	Defines a custom sequence of marks and gaps.

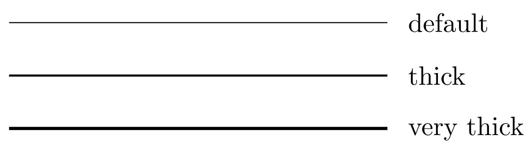
Place line options immediately after `\draw` or collect them in a reusable style.

4.3 Choosing Line Widths

The predefined widths are sufficient for most diagrams.

```
\begin{tikzpicture}
  \draw (0,0) -- (5,0);
  \draw[thick] (0,-0.7) -- (5,-0.7);
  \draw[very thick] (0,-1.4) -- (5,-1.4);
\end{tikzpicture}
```

See



Use the default width for ordinary structure, `thick` for important structure, and `very thick` only for deliberate emphasis.

4.3.1 Exact Widths

An exact width is useful when a publication specification requires one or when several closely controlled widths must be compared.

```
\begin{tikzpicture}
  \draw[line width=0.4pt] (0,0) -- (5,0);
  \draw[line width=0.8pt] (0,-0.7) -- (5,-0.7);
  \draw[line width=1.4pt] (0,-1.4) -- (5,-1.4);
\end{tikzpicture}
```

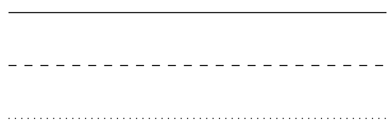
Do not use an exact measurement merely to imitate `thick`. The named options are easier to remember and keep consistent.

4.4 Dashed and Dotted Lines

Pattern can distinguish a secondary relationship without making it heavier.

```
\begin{tikzpicture}
  \draw (0,0) -- (5,0);
  \draw[dashed] (0,-0.7) -- (5,-0.7);
  \draw[dotted] (0,-1.4) -- (5,-1.4);
\end{tikzpicture}
```

See



Common conventions include dashed auxiliary lines, dotted guides, and solid main structure. A convention is useful only when it is applied consistently.

4.4.1 Combining Width and Pattern

Several options may appear in one list:

```
\draw[thick,dashed] (0,0) -- (5,0);
\draw[very thick,dotted] (0,-0.8) -- (5,-0.8);
```

The comma is essential. Each option changes a different visual property.

4.5 Custom Dash Patterns

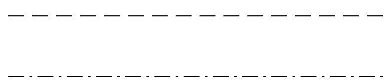
The predefined `dashed` option should be the first choice. When it does not express the required distinction, define the marks and spaces explicitly.

```
\draw[
  dash pattern=on 6pt off 3pt
] (0,0) -- (5,0);
```

A more complex pattern can alternate long and short marks:

```
\draw[
  dash pattern=on 6pt off 2pt on 1pt off 2pt
] (0,0) -- (5,0);
```

See



Common Mistake

Do not create several nearly indistinguishable dash patterns. If readers must study the legend to tell two lines apart, the visual system is too subtle.

4.6 Semantic Line Styles

Low-level options become more useful when they are assigned names based on their purpose.

```
\tikzset{
  main edge/.style={thick},
  auxiliary edge/.style={dashed},
  emphasized edge/.style={very thick}
}
```

The drawing can now describe meaning rather than formatting details:

```
\draw[main edge] (A) -- (B);
\draw[auxiliary edge] (A) -- (C);
\draw[emphasized edge] (B) -- (D);
```

Editing one definition changes every edge with that role.

4.7 Worked Project: A Visual Hierarchy

The project uses the same four named nodes throughout. Only the line system is developed in stages.

4.7.1 Stage 1: Define the Roles

```
\tikzset{
  main edge/.style={thick},
  guide edge/.style={dotted},
  focus edge/.style={very thick,dashed}
}
```

4.7.2 Stage 2: Place the Nodes

```
\node[draw] (A) at (0,2) {$A$};
\node[draw] (B) at (4,2) {$B$};
\node[draw] (C) at (0,0) {$C$};
\node[draw] (D) at (4,0) {$D$};
```

4.7.3 Stage 3: Draw by Role

```
\draw[main edge] (A) -- (B);
\draw[main edge] (A) -- (C);
\draw[guide edge] (C) -- (D);
\draw[focus edge] (B) -- (D);
\draw[guide edge] (A) -- (D);
```

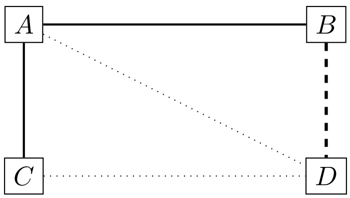
4.7.4 The Complete Picture

```
\begin{tikzpicture}
  \tikzset{
    main edge/.style={thick},
    guide edge/.style={dotted},
    focus edge/.style={very thick,dashed}
  }

  \node[draw] (A) at (0,2) {$A$};
  \node[draw] (B) at (4,2) {$B$};
  \node[draw] (C) at (0,0) {$C$};
  \node[draw] (D) at (4,0) {$D$};

  \draw[main edge] (A) -- (B);
  \draw[main edge] (A) -- (C);
  \draw[guide edge] (C) -- (D);
  \draw[focus edge] (B) -- (D);
  \draw[guide edge] (A) -- (D);
\end{tikzpicture}
```

See



Design

The source records three meanings: main, guide, and focus. The style definitions translate those meanings into width and pattern.

4.8 Debugging Workshop

4.8.1 Options After the Path

Incorrect:

```
\draw (0,0) -- (4,0) [dashed];
```

Correct:

```
\draw[dashed] (0,0) -- (4,0);
```

4.8.2 Missing Commas

Incorrect:

```
\draw[very thick dashed] (0,0) -- (4,0);
```

Correct:

```
\draw[very thick,dashed] (0,0) -- (4,0);
```

4.8.3 Changing Style to Repair Geometry

Thickness and pattern cannot move a path. If a line is in the wrong place, inspect its endpoints. If it has the wrong emphasis, inspect its style.

4.8.4 Too Many Competing Styles

Reduce the number of distinct widths and patterns until every remaining difference has a clear purpose.

4.9 Exercises

1. Draw the same segment with the default, `thick`, and `very thick` widths.
2. Compare exact widths of `0.4pt`, `0.8pt`, and `1.4pt`.
3. Draw one solid, one dashed, and one dotted path.
4. Combine `thick` with `dashed`.
5. Create one restrained custom dash pattern.
6. Define styles named `main edge` and `auxiliary edge`.
7. Apply both styles to a small diagram without moving any node.
8. Revise the worked project so that a different edge receives emphasis.

4.10 Solutions

4.10.1 Exercises 1–4

```
\draw (0,0) -- (5,0);
\draw[thick] (0,-0.7) -- (5,-0.7);
\draw[very thick] (0,-1.4) -- (5,-1.4);
```

```
\draw[line width=0.4pt] (0,0) -- (5,0);
\draw[line width=0.8pt] (0,-0.7) -- (5,-0.7);
\draw[line width=1.4pt] (0,-1.4) -- (5,-1.4);
```

```
\draw (0,0) -- (5,0);
\draw[dashed] (0,-0.7) -- (5,-0.7);
\draw[dotted] (0,-1.4) -- (5,-1.4);
```

```
\draw[thick,dashed] (0,0) -- (5,0);
```

4.10.2 Exercises 5–7

```
\draw[dash pattern=on 5pt off 2pt]
  (0,0) -- (5,0);
```

```
\tikzset{
  main edge/.style={thick},
  auxiliary edge/.style={dashed}
}
```

```
\draw[main edge] (A) -- (B);
\draw[auxiliary edge] (A) -- (C);
```

For Exercise 8, exchange the `focus edge` style assignment with one of the other edges; the coordinates should remain unchanged.

4.11 ChatGPT Workshop

Ask ChatGPT to keep the geometry fixed while evaluating the visual hierarchy.

1. Keep every coordinate unchanged and suggest a restrained line-style hierarchy for main, auxiliary, and emphasized edges.
2. Refactor these repeated width and pattern options into semantic TikZ styles.
3. Explain whether each dashed or dotted line in this diagram has a clear purpose.
4. Check only the line-style options and commas in this TikZ picture. Do not alter its geometry.
5. Compare the predefined `dashed` option with one simple custom dash pattern and recommend the clearer choice.

4.12 Final Checklist

Before continuing to Chapter 5, check that you can

1. distinguish line geometry from line appearance;
2. choose among the default, `thick`, and `very thick`;
3. set an exact width when it is genuinely required;
4. use solid, dashed, and dotted patterns consistently;
5. combine width and pattern options with commas; and
6. define semantic styles for repeated line roles.